WORKED SOLUTION

STEP III 2009, Question 12

Probability and Statistics • Probability Generating Functions

STEP III 2009, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q12

The key idea

In part (i) each round of tossing halves the expected number of coins, but the number of tosses is itself random, so we condition on it: given the previous count, the next count is binomial and its mean is easy. That turns the expectations into a geometric sequence. In part (ii) the total is a sum of independent, identically distributed pieces, and a probability generating function turns "sum of independent variables" into "product of generating functions".

Solution

Part (i)

Each toss is a head with probability $\frac{1}{2}$, independently, so $X_1 \sim \mathrm{B}(k, \frac{1}{2})$ and

$$\mathrm{E}(X_1) = \tfrac{1}{2}k.$$

Given $X_1 = x_1$, Albert makes x_1 fresh, independent tosses, so $X_2 \mid X_1 = x_1 \sim \mathrm{B}(x_1, \frac{1}{2})$ and

$$\mathrm{E}(X_2 \mid X_1 = x_1) = \tfrac{1}{2}x_1.$$

(If $x_1 = 0$ he makes no tosses and $X_2 = 0$, which agrees.) Averaging over the possible values of X_1, by the law of total expectation,

$$\mathrm{E}(X_2) = \sum_{x_1=0}^{k} \mathrm{E}(X_2 \mid X_1 = x_1)\,\mathrm{P}(X_1 = x_1) = \tfrac{1}{2}\sum_{x_1=0}^{k} x_1\,\mathrm{P}(X_1 = x_1) = \tfrac{1}{2}\,\mathrm{E}(X_1) = \tfrac{1}{4}k,$$

as required.

? **Ask yourself** Why can't we just say "X_2 is $\mathrm{B}(\frac{1}{2}k, \frac{1}{2})$"?

Because the number of tosses in the second round is the random variable X_1, not the number $\frac{1}{2}k$; indeed $\frac{1}{2}k$ need not even be an integer. Conditioning on X_1 fixes the number of tosses, and only then is the binomial mean available.

Exactly the same argument works at every stage: given $X_i = x_i$, $X_{i+1} \sim \mathrm{B}(x_i, \frac{1}{2})$, so $\mathrm{E}(X_{i+1}) = \frac{1}{2}\mathrm{E}(X_i)$. By induction,

$$\mathrm{E}(X_i) = \frac{k}{2^i} \qquad (i \geq 1),$$

and summing the geometric series (ratio $\frac{1}{2}$),

$$\sum_{i=1}^{\infty} \mathrm{E}(X_i) = k\left(\tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{8} + \cdots\right) = k.$$

Part (ii)

For one coin, $Y_i = n$ means n heads followed by a tail, so

$$\mathrm{P}(Y_i = n) = \left(\tfrac{1}{2}\right)^n \cdot \tfrac{1}{2} = \left(\tfrac{1}{2}\right)^{n+1}, \qquad n = 0, 1, 2, \ldots$$

Its probability generating function is a geometric series, convergent for $|t| < 2$:

$$G_{Y_i}(t) = \sum_{n=0}^{\infty} \left(\tfrac{1}{2}\right)^{n+1} t^n = \frac{\frac{1}{2}}{1 - \frac{1}{2}t} = \frac{1}{2 - t}.$$

The k coins are tossed separately, so $Y_1, \ldots, Y_k$ are independent, and the generating function of a sum of independent variables is the product of their generating functions. Hence

$$G_Y(t) = \mathrm{E}(t^{Y_1 + \cdots + Y_k}) = \prod_{i=1}^{k} \mathrm{E}(t^{Y_i}) = (2 - t)^{-k}.$$

Mean and variance. Differentiating,

$$G_Y'(t) = k(2 - t)^{-k-1}, \qquad G_Y''(t) = k(k+1)(2 - t)^{-k-2},$$

so $G_Y'(1) = k$ and $G_Y''(1) = k(k+1)$. Therefore

$$\mathrm{E}(Y) = G_Y'(1) = k,$$

and, since $G_Y''(1) = \mathrm{E}(Y(Y-1))$,

$$\mathrm{Var}(Y) = G_Y''(1) + G_Y'(1) - (G_Y'(1))^2 = k(k+1) + k - k^2 = 2k.$$

The distribution. $\mathrm{P}(Y = r)$ is the coefficient of t^r in $G_Y(t)$. Write

$$G_Y(t) = 2^{-k}\left(1 - \tfrac{1}{2}t\right)^{-k},$$

and use the binomial series for a negative index (valid for $|t| < 2$):

$$(1-x)^{-k} = \sum_{r=0}^{\infty} \binom{k+r-1}{r} x^r.$$

(The coefficient is $\dfrac{k(k+1)\cdots(k+r-1)}{r!}$, which is $\binom{k+r-1}{r}$.) With $x = \frac{1}{2}t$,

$$\mathrm{P}(Y=r) = \binom{k+r-1}{r} \frac{1}{2^{k+r}}, \qquad r = 0, 1, 2, \ldots$$

Ask yourself Is there a way to see this formula directly?

Yes. Think of all the tosses in one long line: $Y = r$ means that the kth tail appears on toss number $k+r$. That particular sequence of $k+r$ tosses has probability $2^{-(k+r)}$, and the last toss must be a tail, so we only choose which r of the first $k+r-1$ tosses are heads: $\binom{k+r-1}{r}$ ways. This is the negative binomial distribution.

Checks

- With $k = 1$, $\mathrm{P}(Y=r) = 2^{-(r+1)}$, the single-coin geometric distribution, and $\mathrm{E}(Y) = 1$, $\mathrm{Var}(Y) = 2$, the familiar values for a geometric variable with $p = \frac{1}{2}$ counted from 0.
- $G_Y(1) = 1$, as every generating function must satisfy.
- With $k = 2$, $\mathrm{P}(Y=0) = \frac{1}{4}$ (both first tosses tails) and $\mathrm{P}(Y=1) = \binom{2}{1}\frac{1}{8} = \frac{1}{4}$ (head then tail on one coin, tail on the other: two orders, each $\frac{1}{8}$).
- Exact computation with fractions confirmed $\mathrm{E}(X_i) = k/2^i$ for $k \le 7$ and $i < 40$ (so the partial sums approach k), and the formula for $\mathrm{P}(Y=r)$, the mean k and variance $2k$ for $k \le 6$.

Common slips

- Taking $\mathrm{P}(Y_i = n) = (\frac{1}{2})^n$, forgetting the final tail; the probabilities then sum to 2.
- Using $\mathrm{Var}(Y) = G''(1) - \big(G'(1)\big)^2$, which misses the $G'(1)$ term and gives k instead of $2k$.
- Writing the coefficient as $\binom{k+r}{r}$: the last toss is fixed as a tail, so only $k+r-1$ tosses are free.

Beyond STEP

Part (i) is a branching process: each "individual" (a head) has a random number of "offspring" in the next generation, here $\mathrm{B}(1, \frac{1}{2})$, and since the mean number of offspring

is less than 1 the population dies out. The conditioning step $\mathrm{E}(X_{i+1}) = \mathrm{E}\big(\mathrm{E}(X_{i+1} \mid X_i)\big)$ is the tower property, and in a first-year course the generating function of X_i is shown to be an i-fold composition of the offspring generating function. Grimmett and Welsh, *Probability: An Introduction*, treats both generating functions and branching processes.

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